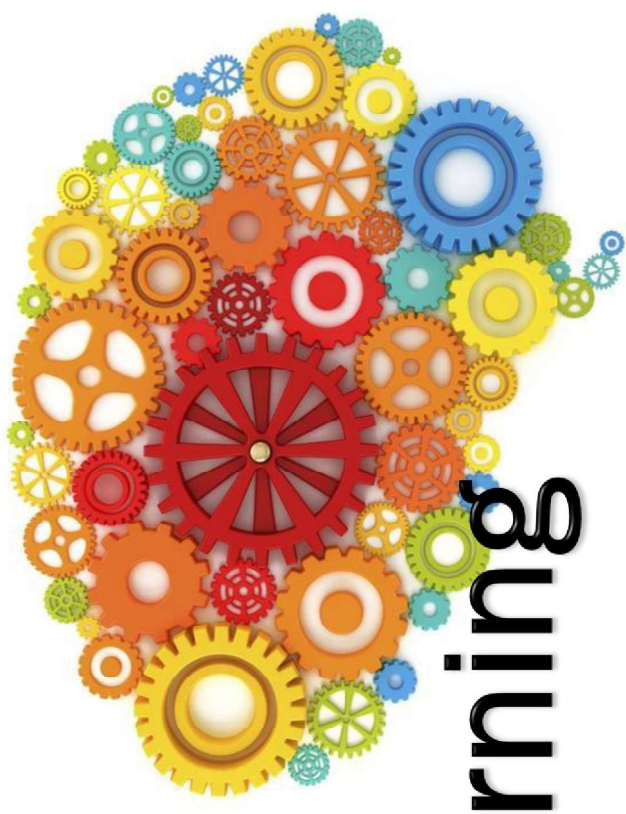




Nonverbal Learning Disability

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Disclosure

- ▶ I have no actual or potential conflict of interest in relation to this program/presentation.



Agenda

- ▶ Where does NLD stand in the field of psychology? Domestic and abroad
- ▶ Define NLD
- ▶ Understand theories of NLD neurobiology
- ▶ In depth look at cognitive-emotional profile
- ▶ Recommendations for home and school



My Experience with NLD

- ▶ “Accidental Expert”



Nonverbal Learning Disorder (NLD): The Basics

- ▶ NLD is not accepted or recognized in the US
- ▶ NLD is fairly accepted, but not formally recognized, throughout the rest of the world
- ▶ Britain and Australia have published extensively on NLD
- ▶ However, there is no diagnostic code in the DSM-IV, DSM-V, ICD-9, or ICD-10



Diagnostic Coding

- ▶ With the DSM-V, PDD NOS is no longer.
- ▶ Options (depending on client's symptom profile)
 - DSM-IV
 - 299.80 Pervasive Developmental Disorder NOS
 - DSM-V
 - 315.39 Social (Pragmatic) Communication Disorder
 - ICD-10
 - F84.1 Atypical Autism
 - F84.8 Other Pervasive Developmental Disorders
 - F84.9 Pervasive Developmental Disorder, Unspecified



Brief History of Learning Disorders

- ▶ “Learning disorders” coined in 1960s
 - Defined as children with normal intelligence who were not achieving adequately in the usual educational settings
- ▶ 2 subtypes: Speech or Achievement
- ▶ It became apparent that many children did not fit into these categories but continued to struggle
- ▶ These children struggled with many aspects of their environment. They lacked social perception (“reading between the lines”)



Outward Appearance

- ▶ Physically awkward (e.g., don't quite know what to do their hands during casual conversation; prone to show anxiety-induced and potentially embarrassing behaviors in public),
- ▶ Socially intrusive (e.g., stands too close or follows someone around during casual conversation; not sure when or how to join a conversation)
- ▶ Socially isolated (e.g., unsure or unaware of their option to join a group or initiate social interaction).
- ▶ Incessant talking (narrate their thoughts aloud, sometimes sharing too much detail without realizing, or repeat details and have trouble getting to the point or sharing the "big picture" during conversation)



Prevalence??

- ▶ “Nonverbal learning disabilities affect one in ten learning disabled children”

Liza Little, PsyD, RN. "The Misunderstood Child: The Child With a Nonverbal Learning Disorder".



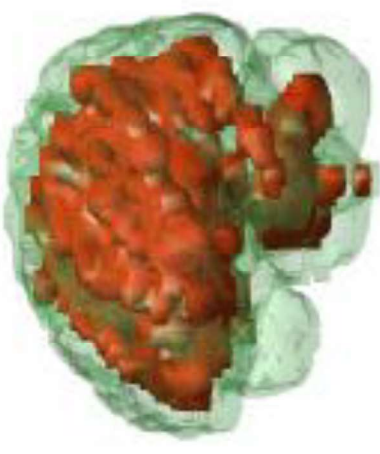
NLD Misunderstood

- ▶ Because NLD does not formally exist, professionals have made many attempts to label these individuals
- ▶ Often mistaken for, or for a lack of a better option...
 - Asperger's Syndrome
 - PDD, NOS
 - ADHD
 - Math or Reading Disorder



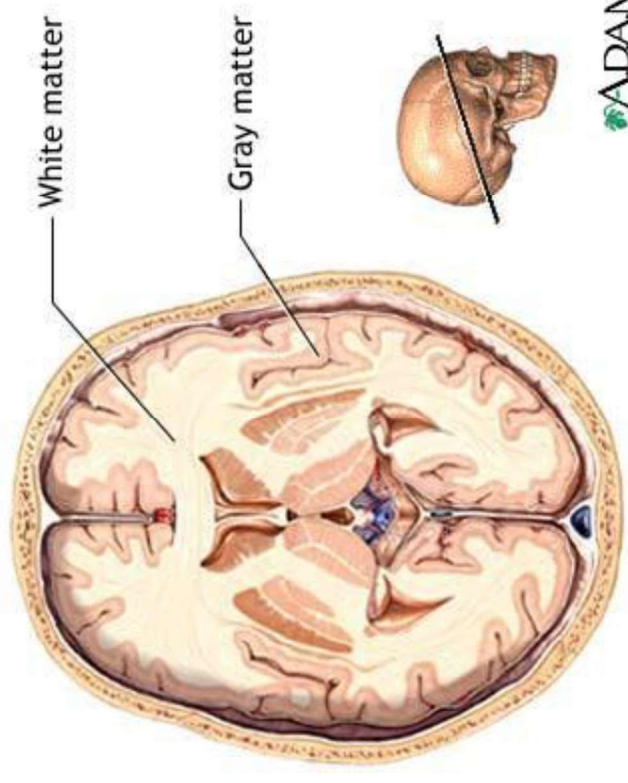
Neurobiology of NLD

- ▶ Neuro-developmental disorder of the RIGHT hemisphere of the brain that is present at or soon after birth.
 - There is a heavily researched connection between hydrocephalus and spina bifida
- ▶ Byron Rourke's White Matter Hypothesis that NLD...
 - involves destruction/dysfunction of white matter
 - reduction of fibers that connect the two hemispheres



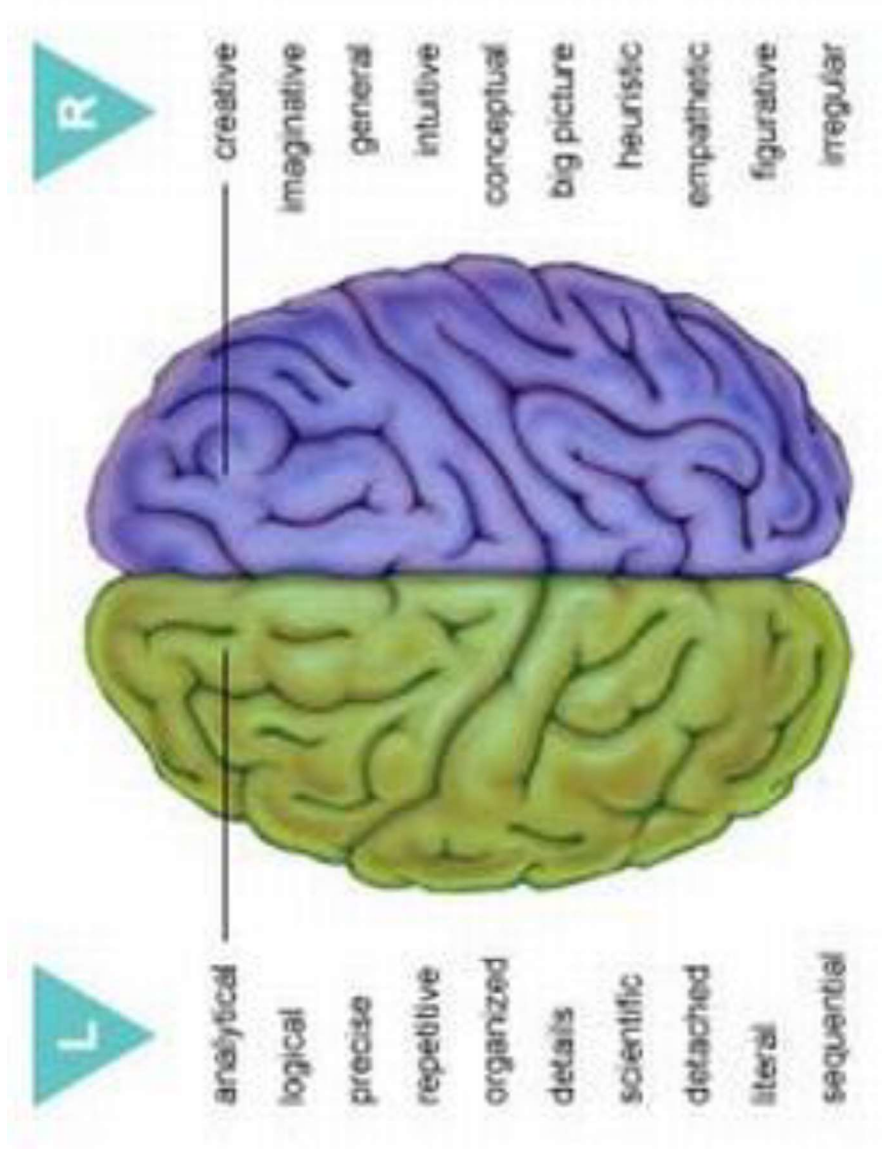
White Matter Hypothesis of NLD

- ▶ The more dysfunctional/damaged the white matter is, the greater the chance of NLD
- ▶ The developmental stage and type of white matter damage sustained has significant influence on manifestation of NLD
- ▶ The integrity of right hemisphere white is crucial for developing and preserving certain functions and damage will hinder learning that requires the integration of complex information



Lateralization of Brain

- Both hemispheres are not equally good at all tasks



Lateralization of Brain

- ▶ Ratio of grey-white matter is higher in the left hemisphere than in the right
- ▶ Organization of the left hemisphere, relative to the right, emphasizes process or transfer within regions rather than transfer across regions



Right Hemisphere Damage

- ▶ Attention disorders
- ▶ Visual-perceptual disorders
- ▶ Memory disorders
- ▶ Affective and emotional difficulties

[All of which are typically present in NLD]



NLD “Diagnostic Criteria”

- ▶ Deficits in 3 major areas
- ▶ Motoric: lack of coordination, problems with balance and graphomotor skills
- ▶ Visual-spatial-organizational: lack of image, poor visual recall, faulty spatial perception, and difficulty with spatial relations
- ▶ Social: inability to comprehend nonverbal communication, difficulty adjusting to transitions and novel situations, and deficits in social judgment



Unique Assets

- ▶ These children have unique assets!
 - Early speech and vocabulary development
 - Remarkable rote memory skills
 - Attention to detail
 - Early development of reading skills and excellent spelling skills
 - Eloquent verbal ability
 - Strong auditory retention



Unique Challenges

- ▶ NLD features of change (for the worse!) as a child gets older.
- ▶ For example, a first grader with NLD may demonstrate very strong verbal skills ('a little professor') and be expected to know how to apply these skills to promote school success.
- ▶ Over time, given weaknesses in such areas as organization, abstract thinking and social cueing, in conjunction with apparent early strengths in isolated skill areas, this same student might quickly fall behind and be perceived as not trying hard enough.

Difficulties in Depth

- ▶ Nonverbal Functioning
 - Perceptual visual deficits
 - Non-verbal problem solving, concept formation and hypothesis testing
- ▶ Academic functioning
- ▶ Social-Emotional Functioning
 - Social relationships
 - Emotional development



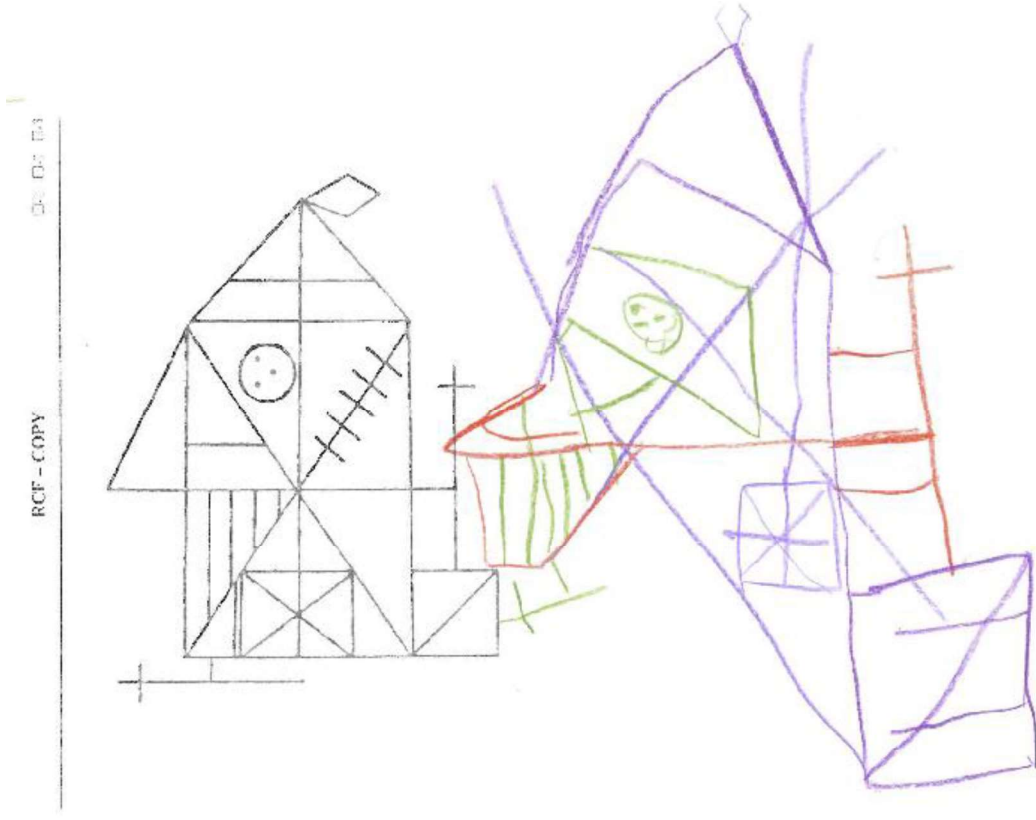
Nonverbal Functioning

- ▶ Perceptual visual deficits
 - Difficulty with complex tasks that involve both thought and motor activity.
 - Kids do well with established routines but have noticeable problems with new situations or novel material.



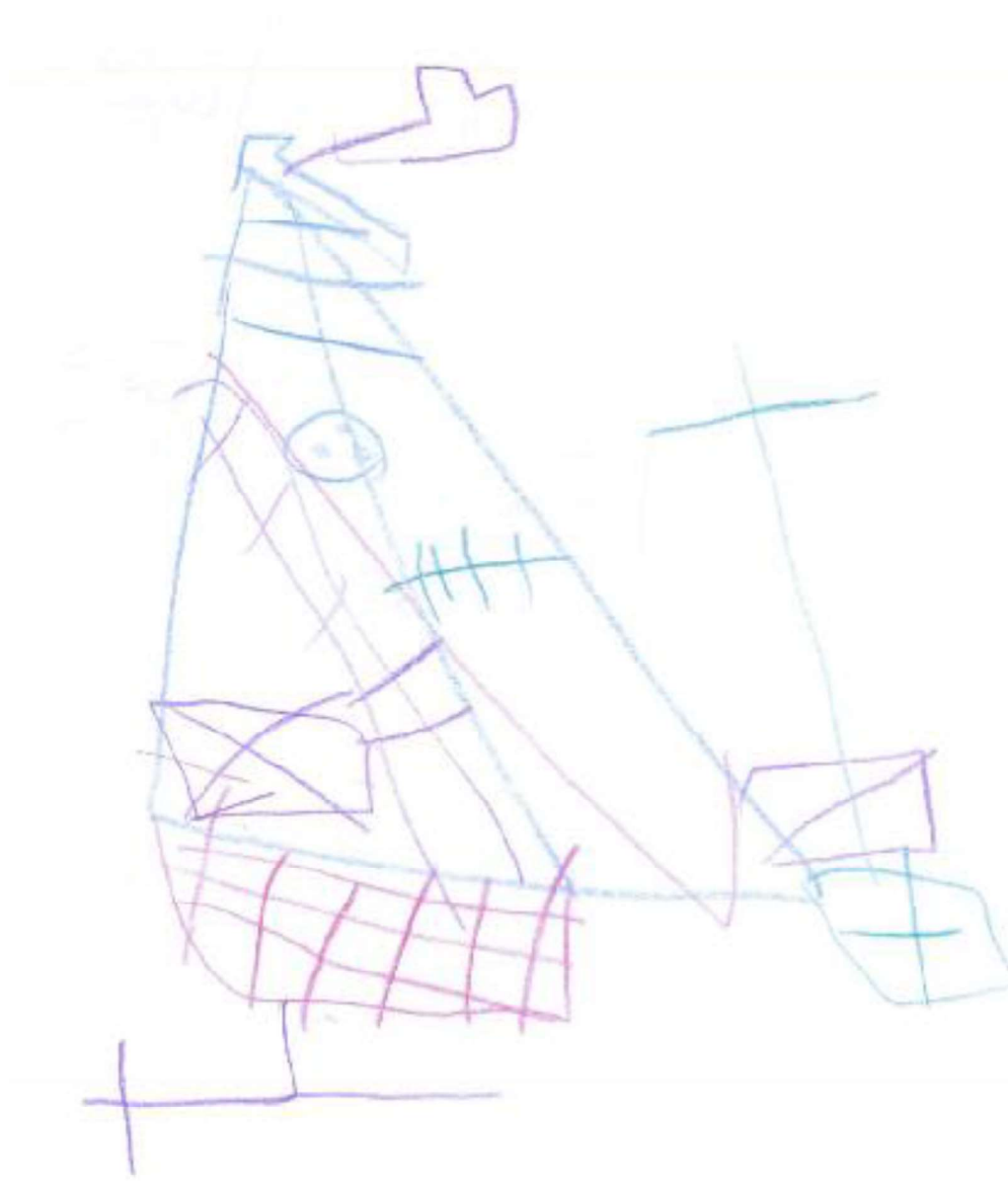
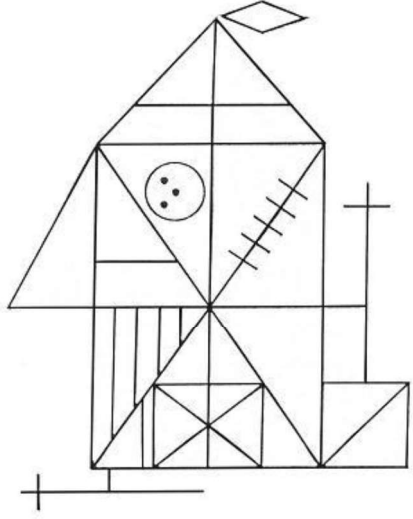
REY-O Examples

8 year old male



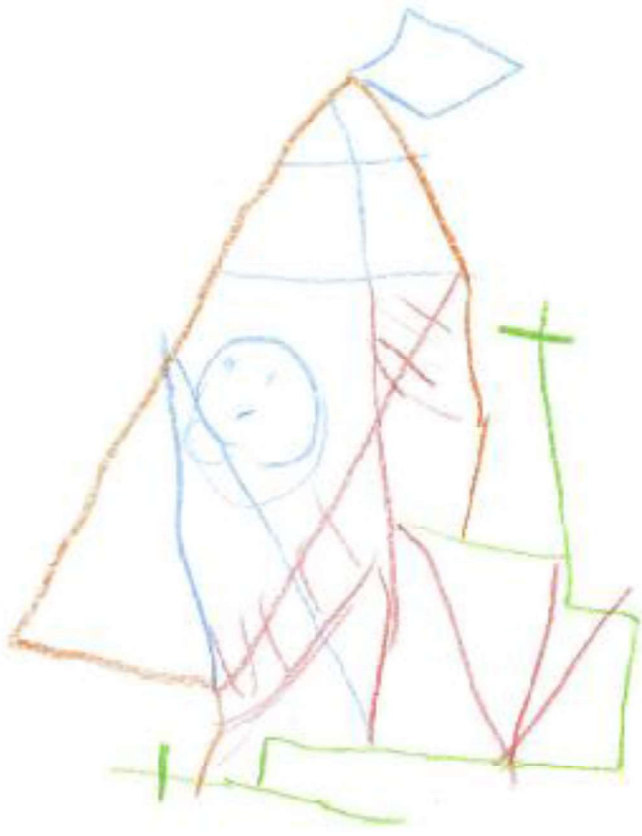
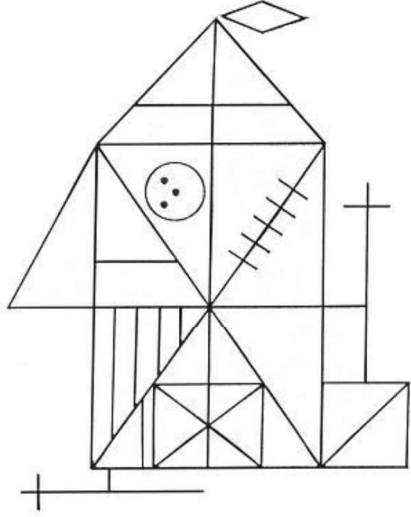
REY-O Examples

12 year old male



REY-O Examples

14 year old female

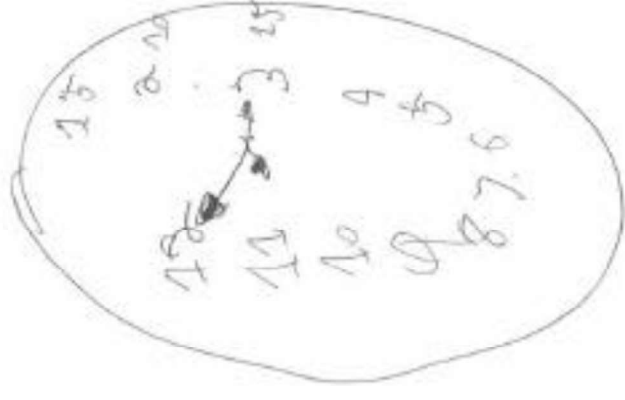


Clock Drawing Example

Attempt #1



Attempt #2



9 year old female

Nonverbal Functioning

- ▶ Non-verbal problem solving, concept formation and hypothesis testing
 - Deficits in dealing with cause and effect thinking
 - Poor appreciation of incongruities and adapting to novel and complex situations
 - Reliance on over-learned behaviors



Academic Functioning

- ▶ Word decoding and spelling come easy for rote material
- ▶ Early motor and motor planning difficulties
- ▶ Math calculation and discrimination of symbols is problematic
- ▶ Reading comprehension for inferential information is limited
- ▶ AUDITORY LEARNERS!



Social-Emotional Functioning

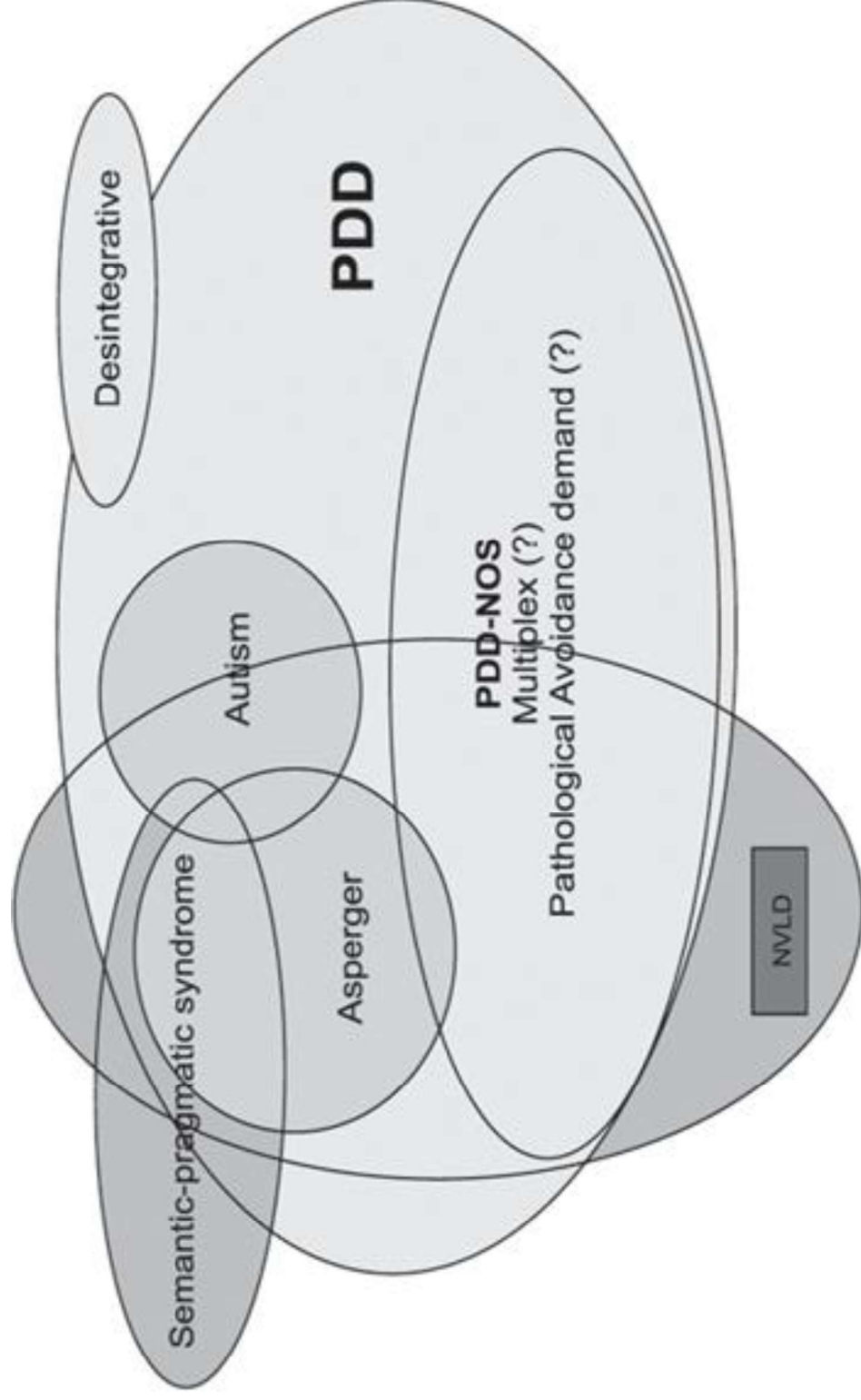
- ▶ Social Relationships
 - Misinterpret body language and tone of voice
 - Do not perceive subtle environmental cues
 - Reactions, personal space
- ▶ Emotional Development
 - Vulnerable to internalizing symptoms such as depression, withdrawal, and anxiety
 - Peer teasing



Differential Diagnosis

NLD	ADHD
Difficulty with spatial relationships and perceptions; frequently bumps into objects; may have difficulty with maintaining balance in seat	Often fidgets or squirms; difficulty remaining still or seated when sustained visual attention is required
Slow motoric performance on non-verbal tasks with hypervigilance to details vs big picture	Easily distracted and impulsive; poor planning and follow-through with details
Talkative; reliant on verbal mediation; may not be aware of manipulation or deception	Talks excessively; impatient and often loses things; may be manipulative and deceptive
Poor social judgment; frequent avoidance of novel situations	Seeks out novelty with enthusiasm; risk-taking behaviors
Comorbidity with depressive or anxious symptoms	Comorbidity with oppositional defiant behaviors

Differential Diagnosis



Differential Diagnosis

Nonverbal Learning Disorder	Asperger's Syndrome	Pervasive Developmental Disorder
Early speech and vocabulary	No language delay	Speech develops, then plateaus
Poor pragmatics and prosody	Poor pragmatics and prosody	
Normal to superior IQ; VIQ>PIQ; verbal can be notably developed	Normal to superior IQ; VIQ>PIQ	Normal IQ, then fails to maintain commensurate with peers
Significant gross, fine and visual-motor problems; dysphasia	Clumsiness; gross motor is generally WNL; fine motor is poor	Coordination develops normally and generally remains WNL
Lacking verbal communication skills (receptive and expressive); naïve; poor social judgment	Perseverating behaviors; restrictive areas of interest; hyperfocus on details	Restrictive; repetitive; stereotypical patterns of behavior



NLD vs Asperger's Further...

- ▶ Studies conducted by the Yale Child-Study Group suggest that up to 80% of children who meet the criteria for Asperger's also present with features suggesting NLD.
- ▶ There are no studies on overlap in the other direction, most likely children with the more severe forms of NLD also meet criteria for Asperger's.
- ▶ Children from both groups are socially awkward and pay over-attention to detail and parts, while missing main themes or underlying principles.
- ▶ However, the two groups differ in the range of severity. Professionals reserve an Asperger's diagnosis for children with more severe social impairment and behavioral rigidity; some symptoms may overlap with high functioning autism

Assessment of NLD

- ▶ It is not simply a matter of Performance IQ being less than Verbal IQ, since there may be many reasons for such a discrepancy besides NLD.
- ▶ NLD can be present even if no discrepancies between strong verbal ability and poor performance show up.
- ▶ One does not need to have every characteristic of NLD in dramatic form.
- ▶ NLD can be complicated by an array of psychiatric and social/familial problems
- ▶ It is important to assess the whole child-world system, not just the cognitive status.

Neuropsychological Profile

- ▶ VIQ > PIQ by at least 10 points
- ▶ Poor performance on tests that time eye/hand, motor planning and anticipation
- ▶ Reading scores significantly better than arithmetic
- ▶ Poor mimicking of hand patterns
- ▶ Very poor finger agnosia – unable to identify letters drawn onto fingers
- ▶ Very poor stereognosis – unable to identify objects placed in hand with vision occluded.
- ▶ Normal to superior grip strength, but poor fine motor dexterity
- ▶ Significant praxis deficits (motor planning, ideation and execution), pronounced on left side
- ▶ Poor sentence memory and very poor phonemically cued verbal fluency.
- ▶ Articulation and enunciation is good, poor content of speech, poor use of pragmatics

Special Education Dilemma

- ▶ The field of psychology does not recognize NLD
- ▶ Schools cannot be expected to either
- ▶ How does a child with NLD qualify for support when they do not fit into one the boxes?
 - Cognitively Impaired
 - Emotionally Impaired
 - Speech and Language Impaired
 - Specific Learning Disability
 - Autism Spectrum Disorder
 - Other Health Impairment



Special Education Dilemma

- ▶ Children with NLD can benefit from interventions within each support category
- ▶ Speech and Language Impaired
 - Improve pragmatic language
- ▶ Specific Learning Disorder
 - Reading comprehension and math assistance
- ▶ Autism Spectrum Disorder
 - Social skills development
- ▶ Other Health Impairment
 - Accommodations and/or OT for sensory, attention, visual perceptual, gross/fine motor abilities

Ideal Classroom

- ▶ A small school in terms of the actual facility, student population and class size
- ▶ A teacher and paraprofessional/aide in the classroom
- ▶ A limited number of teachers and class changes
- ▶ Continuity of teachers and students
- ▶ A high level of structure, predictability, and routine
- ▶ A caring staff, accepting students, and zero tolerance for bullying
- ▶ Reduced visual and auditory distractions
- ▶ Supervision and assistance for social and spatial challenges
- ▶ A nurturing environment for the little ones, weaning to supportive, rather than independent, for the older students.



School Specific Ideas

- ▶ Parent, buddy or teacher assistant to serve as a scribe
- ▶ Reduction of tasks requiring folding, cutting, or arranging materials in a visual spatial manner
- ▶ Typing skills to facilitate note taking/writing
- ▶ Teach verbal mediation (self-talk)
- ▶ Reading out loud improves comprehension
- ▶ Chunking of information



School Specific Ideas

- ▶ To teach math, use real life examples; heavy on verbal explanation and hands on experiences
- ▶ Use graph paper
- ▶ Frequent breaks
- ▶ Extra time and/or buddy to get from class to class
- ▶ Encourage generalization of learned concepts (overlearning)
- ▶ Build opportunities for leadership in small groups



Intervention Ideas for the Family

- ▶ Parent education
 - ▶ Occupational therapy to develop fine motor and visual motor integration skills
 - ▶ Early speech/language therapy in very young children that may have difficulty with the motor planning involved in speech production
 - ▶ Social skills groups to cultivate social awareness, teach nonverbal behavior, encourage self-evaluation and promote specific problem-solving strategies
 - ▶ CBT to develop coping techniques
- 

Intervention Ideas for the Family

- ▶ To improve eye-hand coordination and problem solving skills
 - Computer games and handheld electronic toys.
 - Skills with these types of activities may also prove socially beneficial, as the child's skill will be prized by his or her peer group.

Intervention Ideas for the Family

- ▶ To improve gross motor skills, physical coordination, and dexterity
 - swimming, archery, canoe/kayak, dog obedience, in-line skating, hiking (with good boots and a walking stick), art, and music.
 - It is wise to begin with private lessons, and if the child develops a basic level of proficiency, then small group lessons may be socially beneficial.
 - Avoid team sports, since children with NLD have a great deal of difficulty with the rules of the game and the complexity of the social interaction involved.
 - The goal is to build competence AND confidence.

Intervention Ideas for the Family

- ▶ Structured organizations (such as scouts, 4-H, church youth groups, etc.) are generally the safest social outlets for children with NLD.
- ▶ However, make sure that the facilitator is aware of the child's strengths and weaknesses so that he/she is not left to socially fend for themselves.

NLD Resources

- ▶ Online
 - www.nldonline.com
 - www.nldontheweb.org
 - www.ldonline.org
 - U of Michigan
 - <http://www.med.umich.edu/yourchild/topics/nld.htm>
 - Lake Michigan Academy



NLD Resources

► Books

- Sue Thompson's book is the best all round resource. It is now titled The Source for Nonverbal Learning Disorders.
- Kathy Allen, a colleague of Sue's, has self-published Star Shaped Pegs, Square Holes - which helps the child (10-14 yrs.) understand his/her NLD.
- Deborah Green's book, Growing Up With NLD, an autobiographical look at life with a successful adult; also includes research and resources.
- The Nonverbal Learning Disorder Guide for Teachers, Parents, Employers and Therapists by Rondlayn Varney



NLD Resources

► Books

- Unaware: Living With Non-Verbal Learning Disability by Laurie Reed.
- Inside Out: What Makes A Person With Social-Cognitive Deficits Tick? A Manual & Workbook by Michelle Garcia Winner
- Nowicki & Duke have written Teaching Your Child the Language of Social Success and Helping the Child Who Doesn't Fit In
- **Byron Rourke** has two definitive books which provide a clinical look at the NLD syndrome. Syndrome of Nonverbal Learning Disabilities and Nonverbal Learning Disabilities, the Syndrome and the Model



Thank you

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